



**SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)**

Constituent of Symbiosis International (Deemed University), Pune

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**SEM: III**

## ASSIGNMENT No: 5

**TITLE:** Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

### Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

### Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

### Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections

Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects.

These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

### Conclusion:

This experiment demonstrated the practical implementation of ArrayList, Vector, and StringBuffer in Java. ArrayList and Vector were used for dynamic data storage and management, while StringBuffer was used for efficient string manipulation. The developed applications successfully performed add, remove, and display operations, improving the understanding of Java Collections Framework and mutable string handling in real-world programming.

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```
import java.util.ArrayList;

class TodoList {

    public static void main(String[] args) {

        ArrayList<String> tasks = new ArrayList<>();

        tasks.add("Complete Java Assignment");

        tasks.add("Study for Exam");
```

```
tasks.add("Submit Project");

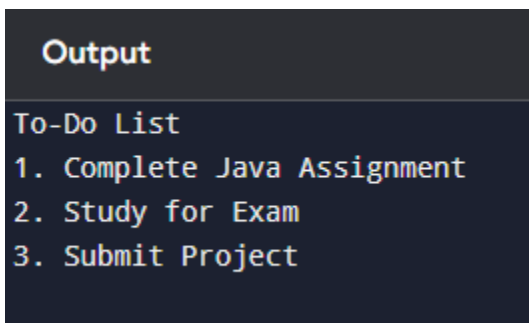
StringBuffer sb = new StringBuffer();

sb.append("To-Do List\n");

for (int i = 0; i < tasks.size(); i++) {
    sb.append((i + 1) + ". " + tasks.get(i) + "\n");
}

System.out.println(sb);
}
```

## OUTPUT

A screenshot of a Java program's output. It shows a dark-themed window with a title bar that says "Output". The content of the window is a text-based To-Do List. It starts with "To-Do List" on the first line, followed by three numbered items: "1. Complete Java Assignment", "2. Study for Exam", and "3. Submit Project".

```
Output
To-Do List
1. Complete Java Assignment
2. Study for Exam
3. Submit Project
```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
import java.util.ArrayList;
```

```
import java.util.Scanner;
```

```
class CourseRegistration {
```

```
    public static void main(String[] args) {
```

```
        ArrayList<String> courses = new ArrayList<>();
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int choice;
```

```
        do {
```

```
            System.out.println("\n1. Add Course");
```

```
            System.out.println("2. Remove Course");
```

```
            System.out.println("3. View Courses");
```

```
            System.out.println("4. Exit");
```

```
            System.out.print("Enter Choice: ");
```

```
            choice = sc.nextInt();
```

```
            sc.nextLine();
```

```
            switch (choice) {
```

```
                case 1:
```

```
                    System.out.print("Enter Course Name: ");
```

```
                    courses.add(sc.nextLine());
```

```
                    System.out.println("Course Added.");
```

```
                    break;
```

```
                case 2:
```

```
System.out.print("Enter Course Name to Remove: ");
String course = sc.nextLine();
if (courses.remove(course))
    System.out.println("Course Removed.");
else
    System.out.println("Course Not Found.");
break;
```

case 3:

```
StringBuffer sb = new StringBuffer();
sb.append("Registered Courses\n");
for (int i = 0; i < courses.size(); i++) {
    sb.append((i + 1) + ". " + courses.get(i) + "\n");
}
System.out.println(sb);
break;
```

case 4:

```
System.out.println("Exiting...");
break;
```

default:

```
System.out.println("Invalid Choice.");
```

```
}
```

```
} while (choice != 4);
```

```
        sc.close();  
    }  
}
```

## OUTPUT

```
Output  
  
1. Add Course  
2. Remove Course  
3. View Courses  
4. Exit  
Enter Choice: 1  
Enter Course Name: java  
Course Added.  
  
1. Add Course  
2. Remove Course  
3. View Courses  
4. Exit  
Enter Choice: 3  
Registered Courses  
1. java
```